

MARKETING DATA ANALYSIS

Customer & Campaign Performance Analytics

Data Analytics Project

Excel | Python | Pandas | NumPy | SQL/PostgreSQL | Power BI

Table of Contents

? Project Overview

? Business Problem

? Business Questions

? Dataset Overview

? Tools & Technology

? End-to-End Analytical Workflow

? Initial Data Exploration – Excel

? Data Cleaning & Preparation – Python

? SQL Business Analysis

? Power BI Dashboard

? Dashboard Analysis & Insights

? Key Findings

? Business Recommendations

? Challenges & Data Limitations

? Skills Demonstrated

? Conclusion

? Appendix – PostgreSQL Data Loading

1. Project Overview

This project analyzes a customer marketing dataset to understand campaign response, customer characteristics, product spending, purchasing channels, online activity, customer value, and the relationship between prior engagement and later campaign response. The project follows the stated analytical workflow: Excel → Python (Pandas & NumPy) → SQL/PostgreSQL → Power BI.

The raw dataset contains 2,240 customer records and 28 fields covering customer demographics, household composition, enrollment date, recency, product spending, purchasing channels, campaign acceptance indicators, response, complaints, and country.

The objective is to turn the raw customer-level data into reliable analytical features, answer defined business questions with SQL, and present the resulting patterns through a three-page Power BI dashboard.

Business relevance The analysis is designed to support decisions around campaign targeting, customer segmentation, product focus, channel strategy, and customer engagement.

2. Business Problem

The business problem is to understand which customers respond to marketing campaigns, what customer characteristics are associated with response and spending, which products and purchasing channels contribute most to activity, and which customers represent higher value.

The available data combines campaign acceptance history with demographics, income, family composition, product spending, purchasing activity, web visits, customer enrollment dates, and geography. This makes it possible to compare customer groups and identify patterns that can inform future targeting.

The analysis can support decisions such as:

- Which campaigns and customer segments deserve greater targeting attention.
- Which products and channels should receive greater marketing focus.
- How previous campaign engagement relates to later response.
- Which customer groups show higher spending and purchase activity.
- Where customer engagement is comparatively strong or weak across countries and enrollment cohorts.

Interpretation note The results describe associations in the available historical data; they should not be interpreted as proof that one customer characteristic or campaign caused another outcome.

3. Business Questions

#	Business question	Primary analytical stage
1	Which marketing campaign generated the highest customer acceptance?	SQL + Power BI
2	What customer characteristics are associated with campaign success?	SQL
3	What factors are associated with higher online purchasing activity?	SQL + Power BI
4	Which customer segments are most responsive to marketing campaigns?	SQL + Power BI
5	Which products generate the highest customer spending?	SQL + Power BI
6	Which purchasing channels are performing best?	SQL + Power BI
7	Are marketing campaigns successfully driving additional purchases?	SQL + Power BI
8	Which customers are most valuable based on spending and purchasing behavior?	SQL + Power BI
9	Does discount-driven purchasing indicate an opportunity or dependency?	SQL + Power BI
10	Which countries show the strongest customer engagement?	SQL + Power BI
11	Which customer characteristics should be considered when targeting future campaigns?	SQL
12	Does customer tenure affect campaign response?	SQL + Power BI

13	Does family size influence customer spending?	SQL + Power BI
14	Does previous campaign engagement predict future campaign response?	SQL + Power BI
15	Does web conversion rate differ between customer segments?	SQL
16	Do newer customers have different purchasing behavior?	SQL + Power BI

4. Dataset Overview

The raw CSV contains 2,240 rows and 28 columns. The notebook confirms the same shape during initial inspection. The data dictionary provides the business meaning of the fields.

Metric	Verified value
Rows	2,240
Columns	28
Duplicate rows	0
Missing income values in raw data	24
Education categories	5
Marital-status categories	8
Countries	8
Raw enrollment date format	DD-MM-YYYY
Data years represented by enrollment date	2012–2014

Raw Column Inventory

Column	Data type	Business meaning
id	int64	Customer's unique identifier
year_birth	int64	Customer's birth year
education	object	Customer's education level
marital_status	object	Customer's marital status
income	float64	Customer's yearly household income
kidhome	int64	Number of children in customer's household
teenhome	int64	Number of teenagers in customer's household
dt_customer	object	Date of customer's enrollment with the company
recency	int64	Number of days since customer's last purchase
mntwines	int64	Amount spent on wine in the last 2 years
mntfruits	int64	Amount spent on fruits in the last 2 years
mntmeatproducts	int64	Amount spent on meat in the last 2 years
mntfishproducts	int64	Amount spent on fish in the last 2 years
mntsweetproducts	int64	Amount spent on sweets in the last 2 years
mntgoldprods	int64	Amount spent on gold in the last 2 years
numdealspurchases	int64	Number of purchases made with a discount
numwebpurchases	int64	Number of purchases made through the company's web site
numcatalogpurchases	int64	Number of purchases made using a catalogue
numstorepurchases	int64	Number of purchases made directly in stores
numwebvisitsmonth	int64	Number of visits to company's web site in the last month
acceptedcmp3	int64	1 if customer accepted the offer in the 3rd campaign, 0 otherwise
acceptedcmp4	int64	1 if customer accepted the offer in the 4th campaign, 0 otherwise
acceptedcmp5	int64	1 if customer accepted the offer in the 5th campaign, 0 otherwise
acceptedcmp1	int64	1 if customer accepted the offer in the 1st campaign, 0 otherwise
acceptedcmp2	int64	1 if customer accepted the offer in the 2nd campaign, 0 otherwise
response	int64	1 if customer accepted the offer in the last campaign, 0 otherwise
complain	int64	1 if customer complained in the last 2 years, 0 otherwise
country	object	Customer's location

5. Data Quality Observations

The notebook explicitly checks missing values, duplicate records, data types, categorical distributions, binary fields, and the final column set before loading the data into PostgreSQL.

Check	Result	Treatment / interpretation
Income missing values	24 in raw data	Filled with the median income (51,381.50)
Duplicate rows	0	No duplicate-row removal was required
Date field	String in raw data	Parsed with format %d-%m-%Y
Web conversion rate	11 nulls after feature creation	Rows with zero monthly web visits produce NaN rather than division by zero
Binary campaign/response/complaint fields	0/1 values	Values were inspected before SQL loading
Year_Birth	Retained initially, then removed	Replaced analytically by derived age

The notebook's final null check shows the engineered dataset has 11 remaining null values, corresponding to the web conversion calculation for customers with no web visits.

6. Tools & Technology

Tool	Purpose verified in project
Excel	Initial exploration stage in the stated workflow; no Excel workbook was included in the supplied project package, so specific Excel techniques are not independently verifiable.
Python	Data loading, cleaning, preprocessing, feature engineering, validation, and PostgreSQL loading.
Pandas	DataFrame manipulation, missing-value handling, date conversion, feature creation, validation, and export/loading preparation.
NumPy	Numerical operation used for the web conversion rate calculation.
SQL / PostgreSQL	Business-question queries and analytical aggregation against the loaded marketing table.
Power BI	Three-page interactive dashboard with KPI cards, charts, slicers, and analytical views.

Evidence limitation No Excel workbook was supplied with the project files. The report therefore preserves Excel as the documented first workflow stage but does not attribute unverified Excel formulas, pivots, or charts to the project.

7. End-to-End Analytical Workflow

The project follows the required logical order. Each stage prepares an output for the next stage.

Stage	Activity	Output
01	Raw Dataset	CSV customer marketing data
02	Excel – Initial Data Exploration	Initial understanding of structure and values; detailed workbook evidence not supplied
03	Python – Data Cleaning	Standardized fields, handled missing income, parsed dates, validated quality
04	Pandas & NumPy – Feature Engineering	Created age, tenure fields, spending, purchases, family size, campaign acceptance, web conversion
05	SQL/PostgreSQL – Business Analysis	Answered campaign, segment, product, channel, engagement, tenure, family and value questions
06	Power BI – Dashboard & Visualization	Presented KPIs, campaign performance, customer analysis, product/channel analysis and slicers
07	Insights & Recommendations	Converted verified patterns into business-oriented observations and recommendations

8. Initial Data Exploration – Excel

Excel is the first stage in the stated project workflow. However, no Excel workbook was included in the supplied project files, so the specific exploration techniques, formulas, filters, pivot tables, or observations performed in Excel cannot be independently verified from the evidence package.

The raw CSV and Notebook provide evidence of the dataset structure that was subsequently inspected in Python: 2,240 records, 28 raw columns, 24 missing income values, zero duplicate rows, and categorical distributions for education, marital status, and country.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	color	Year_Birth	Education	Marital_Status	Income	Kidhom	Teenhom	Dt_Customer	Recenc	MntWine	MntFruit	MntMeatProduct	MntFishProduct	MntSweetProduct	MntGoldProduct	NumDealsPurchase	NumWebPurchase	NumVisits
1	1826	1970	Graduation	Divorced	84835	0	0	16-06-2014	0	189	104	379	111	189	218	1	4	
2	1	1961	Graduation	Single	57091	0	0	15-06-2014	0	464	5	64	7	0	37	1	7	
3	10476	1958	Graduation	Married	67267	0	1	13-05-2014	0	134	11	59	15	2	30	1	3	
4	1386	1967	Graduation	Together	32474	1	1	11-05-2014	0	10	0	1	0	0	0	1	1	
5	5371	1989	Graduation	Single	21474	1	0	08-04-2014	0	6	16	24	11	0	34	2	3	
6	7348	1958	PhD	Single	71691	0	0	17-03-2014	0	336	130	411	240	32	43	1	4	
7	4073	1954	2n Cycle	Married	63564	0	0	29-01-2014	0	769	80	252	15	34	65	1	10	
8	1991	1967	Graduation	Together	44931	0	1	18-01-2014	0	78	0	11	0	0	7	1	2	
9	4047	1954	PhD	Married	65324	0	1	11-01-2014	0	384	0	102	21	32	5	3	6	
10	9477	1954	PhD	Married	65324	0	1	11-01-2014	0	384	0	102	21	32	5	3	6	
11	2079	1947	2n Cycle	Married	81044	0	0	27-12-2013	0	450	26	535	73	98	26	1	5	
12	5642	1979	Master	Together	62499	1	0	09-12-2013	0	140	4	61	0	13	4	2	3	
13	10530	1959	PhD	Widow	67786	0	0	07-12-2013	0	431	82	441	80	20	102	1	3	
14	2964	1981	Graduation	Married	26872	0	0	16-10-2013	0	3	10	8	3	16	32	1	1	
15	10311	1969	Graduation	Married	4428	0	1	05-10-2013	0	16	4	12	2	4	321	0	25	
16	837	1977	Graduation	Married	54809	1	1	11-09-2013	0	63	6	57	13	13	22	4	2	
17	10521	1977	Graduation	Married	54809	1	1	11-09-2013	0	63	6	57	13	13	22	4	2	
18	10175	1958	PhD	Divorced	32173	0	1	01-08-2013	0	18	0	2	0	0	2	1	1	
19	1473	1960	2n Cycle	Single	47823	0	1	23-07-2013	0	53	1	5	2	1	10	2	2	
20	2795	1958	Master	Single	30523	2	1	01-07-2013	0	5	0	3	0	0	5	1	1	
21	2285	1954	Master	Together	36631	0	1	28-05-2013	0	213	9	76	4	3	30	3	5	
22	115	1966	Master	Single	43456	0	1	26-03-2013	0	275	11	68	25	7	7	3	5	
23	10470	1979	Master	Married	40662	1	0	15-03-2013	0	40	2	23	0	4	23	2	2	
24	4065	1976	PhD	Married	49544	1	0	12-02-2013	0	308	0	73	0	0	23	2	5	
25	10968	1969	Graduation	Single	57731	0	1	23-11-2012	0	266	21	300	65	8	44	4	8	
26	5985	1965	Master	Single	33168	0	1	13-10-2012	0	80	1	37	0	1	3	3	2	
27	5430	1956	Graduation	Together	54450	1	1	14-09-2012	0	454	0	171	8	19	32	12	9	
28	8432	1956	Graduation	Together	54450	1	1	14-09-2012	0	454	0	171	8	19	32	12	9	
29	453	1956	PhD	Widow	35340	1	1	29-06-2014	1	27	0	12	0	1	5	2	2	
30	9687	1975	Graduation	Single	73170	0	0	31-05-2014	1	184	174	256	50	30	32	1	5	
31	8890	1971	PhD	Divorced	65808	1	1	30-05-2014	1	155	7	80	13	7	10	3	5	
32	9264	1986	Graduation	Married	79529	0	0	27-04-2014	1	423	42	706	73	197	197	1	4	
33	5824	1972	PhD	Together	34578	2	1	11-04-2014	1	7	0	1	0	0	0	1	1	
34	5794	1974	PhD	Married	46374	0	1	17-03-2014	1	408	0	21	0	0	17	3	7	
35	3068	1990	Graduation	Married	18351	0	0	29-10-2013	1	1	12	9	0	14	7	1	2	
36	7962	1987	PhD	Single	95169	0	0	09-10-2013	1	1285	21	449	106	20	20	1	4	
37	2681	1984	2n Cycle	Married	65370	0	0	01-08-2013	1	71	22	112	138	89	29	1	2	

9. Data Cleaning & Preparation – Python

The Jupyter Notebook shows that Python, Pandas, and NumPy were used to prepare the dataset for SQL analysis.

9.1 Loading and inspection

PYTHON

```
import numpy as np
import pandas as pd
df = pd.read_csv('marketing_data.csv')
df.info()
df.describe()
```

The notebook then standardized column names by stripping whitespace and converting names to lowercase.

PYTHON

```
df.columns = df.columns.str.strip()
df.columns = df.columns.str.lower()
```

9.2 Missing income values

The raw dataset contains 24 missing income values. The notebook fills these values with the median income.

PYTHON

```
df['income'] = df['income'].fillna(df['income'].median())
```

Result The raw-data median income is 51,381.50. After this operation, the income field contains no missing values.

9.3 Date conversion and age / tenure features

PYTHON

```
df['dt_customer'] = pd.to_datetime(
    df['dt_customer'], format='%d-%m-%Y'
)
df['customer_year'] = df['dt_customer'].dt.year
df['customer_month'] = df['dt_customer'].dt.month
df['age'] = df['customer_year'] - df['year_birth']
```

This creates enrollment-year and enrollment-month fields and calculates age relative to the customer enrollment year, matching the historical period of the dataset.

9.4 Spending, purchasing and household features

PYTHON

```
spending_cols = ['mntwines', 'mntfruits',
                 'mntmeatproducts', 'mntfishproducts',
                 'mmtsweetproducts', 'mtgoldprods']
df['total_spend'] = df[spending_cols].sum(axis=1)

purchase_cols = ['numwebpurchases',
                 'numcatalogpurchases', 'numstorepurchases']
df['total_purchase'] = df[purchase_cols].sum(axis=1)

df['total_children'] = df['kidhome'] + df['teenhome']
```

9.5. Campaign and web-engagement features

PYTHON

```
campaign_cols = [
    'acceptedcmp1', 'acceptedcmp2', 'acceptedcmp3',
    'acceptedcmp4', 'acceptedcmp5'
]
df['campaign_acceptance'] = df[campaign_cols].sum(axis=1)

df['web_conversion_rate'] = np.where(
    df['numwebvisitsmonth'] > 0,
    df['numwebpurchases'] / df['numwebvisitsmonth'],
    np.nan
)

df['web_visit_flag'] = (
    df['numwebvisitsmonth'] > 0
).astype(int)
```

The web conversion calculation deliberately avoids division by zero. This leaves 11 null conversion-rate values for customers whose monthly web-visit count is zero.

9.6 Validation before SQL loading

PYTHON

```
df.isnull().sum()
df.duplicated().sum()
df.dtypes
df['education'].value_counts()
df['marital_status'].value_counts()
df['country'].value_counts()
df.head()
```

Validation item	Result
-----------------	--------

Rows	2,240
Raw columns	28
Duplicates	0
Missing income before fill	24
Missing income after fill	0
Remaining engineered nulls	11 web_conversion_rate values
Education categories	5
Marital-status categories	8
Countries	8

9.7 Final schema change and SQL load

PYTHON

```
df = df.drop(columns=['year_birth'])
```

```
df.to_sql(
    'marketing',
    engine,
    if_exists='replace',
    index=False
)
```

The notebook records a successful load into the PostgreSQL database marketing_data and table marketing. After feature engineering and removal of year_birth, the DataFrame contains 36 columns.

The screenshot shows a Jupyter Notebook with two visible code cells. The first cell imports numpy and pandas, reads a CSV file named 'marketing_data.csv', and displays the first 10 rows of the DataFrame. The second cell shows the output of df.info(), indicating 2240 rows and 28 columns.

```
1 import numpy as np
2 import pandas as pd
3 df = pd.read_csv('marketing_data.csv')
4 df
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	...	NumStorePurchases	NumWebVisitsMonth	AcceptedCmp3	AcceptedCmp4	AcceptedCmp5	AcceptedCmp1	AcceptedCmp2	Res
0	1826	1970	Graduation	Divorced	84835.0	0	0	16-06-2014	0	189	...	6	1	0	0	0	0	0	0
1	1	1961	Graduation	Single	57091.0	0	0	15-06-2014	0	464	...	7	5	0	0	0	0	0	1
2	10476	1958	Graduation	Married	67267.0	0	1	13-05-2014	0	134	...	5	2	0	0	0	0	0	0
3	1386	1967	Graduation	Together	32474.0	1	1	11-05-2014	0	10	...	2	7	0	0	0	0	0	0
4	5371	1989	Graduation	Single	21474.0	1	0	08-04-2014	0	6	...	2	7	1	0	0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2235	10142	1976	PhD	Divorced	66476.0	0	1	07-03-2013	99	372	...	11	4	0	0	0	0	0	0
2236	5263	1977	2n Cycle	Married	31056.0	1	0	22-01-2013	99	5	...	3	8	0	0	0	0	0	0
2237	22	1976	Graduation	Divorced	46310.0	1	0	03-12-2012	99	185	...	5	8	0	0	0	0	0	0
2238	528	1978	Graduation	Married	65819.0	0	0	29-11-2012	99	267	...	10	3	0	0	0	0	0	0
2239	4070	1969	PhD	Married	94871.0	0	2	01-09-2012	99	169	...	4	7	0	1	1	0	0	0

2240 rows x 28 columns

```
1 df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 2240 entries, 0 to 2239
Data columns (total 28 columns):
#   Column                Non-Null Count  Dtype
---  -
0   ID                     2240 non-null   int64
1   Year_Birth             2240 non-null   int64
2   Education              2240 non-null   str
3   Marital_Status         2240 non-null   str
4   Income                 2216 non-null   float64
5   Kidhome                2240 non-null   int64
6   Teenhome               2240 non-null   int64
7   Dt_Customer            2240 non-null   str
8   Recency                2240 non-null   int64
9   MntWines               2240 non-null   int64
10  MntFruits              2240 non-null   int64
...
```

Insert a readable screenshot showing the notebook's cleaning, feature-engineering and validation cells.

10. SQL Business Analysis

After preparation, the notebook loads the DataFrame into PostgreSQL as the marketing table. The supplied SQL file contains 16 business questions covering campaign acceptance, customer characteristics, online purchasing, segmentation, product and channel performance, discount behavior, country engagement, tenure, family size, previous campaign engagement, web conversion, and newer-customer behavior.

Result verification note The SQL file contains queries but no saved result tables or screenshots. The numeric results

in this report were reproduced from the cleaned source data using the same query logic, rather than invented or copied from an unavailable SQL-output file.

10.1 Campaign acceptance

Business Question

Which marketing campaign generated the highest customer acceptance?

SQL Approach

Each campaign acceptance flag is summed across customers. The six results are combined and ranked from highest to lowest.

SQL Query

```
SQL
SELECT
    'Campaign 1' AS campaign,
    SUM(acceptedcmp1) AS accepted_customers
FROM marketing

UNION ALL
SELECT 'Campaign 2', SUM(acceptedcmp2) FROM marketing
UNION ALL
SELECT 'Campaign 3', SUM(acceptedcmp3) FROM marketing
UNION ALL
SELECT 'Campaign 4', SUM(acceptedcmp4) FROM marketing
UNION ALL
SELECT 'Campaign 5', SUM(acceptedcmp5) FROM marketing
UNION ALL
SELECT 'Latest Campaign', SUM(response) FROM marketing
ORDER BY accepted_customers DESC;
```

Result

Campaign	Accepted customers
Latest Campaign	334
Campaign 4	167
Campaign 3	163
Campaign 5	163
Campaign 1	144
Campaign 2	30

Business Insight

Within the available data, the Latest Campaign has the highest acceptance count at 334. Among the previous campaigns, Campaign 4 is highest at 167, while Campaign 2 is substantially lower at 30.

Insert a screenshot of the PostgreSQL query and its actual result grid. Keep the SQL readable.

10.2 Customer characteristics

Business Question

What customer characteristics are associated with campaign success?

SQL Approach

Customers are grouped by response. Average age, income, spending, and total purchases are compared between responders and non-responders.

SQL Query

```
SQL
SELECT
    response,
    COUNT(*) AS customers,
    ROUND(AVG(age)::numeric, 2) AS avg_age,
    ROUND(AVG(income)::numeric, 2) AS avg_income,
    ROUND(AVG(total_spend), 2) AS avg_spend,
    ROUND(AVG(total_purchase), 2) AS avg_purchases
FROM marketing
GROUP BY response
ORDER BY response DESC;
```

Result

Response	Customers	Avg age	Avg income	Avg spend	Avg purchases
1 (responded)	334	43.33	60,183.24	987.39	15.37
0 (did not respond)	1,906	44.38	50,845.68	538.93	12.04

Business Insight

Responders have higher average income, spending, and purchase activity than non-responders. Average age is similar between the two groups, so income and purchasing behavior show a stronger descriptive difference in this comparison.

10.3 Segment responsiveness

Business Question

Which customer segments are most responsive to marketing campaigns?

SQL Approach

The query compares response rate across education and marital-status categories. Response rate is calculated as the average of the binary response field multiplied by 100.

SQL Query

```
SQL
SELECT
    education,
    COUNT(*) AS customers,
    SUM(response) AS responses,
    ROUND(100.0 * AVG(response), 2) AS response_rate
FROM marketing
GROUP BY education
ORDER BY response_rate DESC;

-- By marital status
SELECT
    marital_status,
    COUNT(*) AS customers,
    SUM(response) AS responses,
    ROUND(100.0 * AVG(response), 2) AS response_rate
FROM marketing
GROUP BY marital_status
ORDER BY response_rate DESC;
```

Result

Education	Customers	Responses	Response rate
PhD	486	101	20.78%
Master	370	57	15.41%
Graduation	1,127	152	13.49%

2n Cycle	203	22	10.84%
Basic	54	2	3.70%

Business Insight

Education response rates range from 3.70% for Basic to 20.78% for PhD. The very small marital categories Absurd and YOLO show 50% rates but contain only two customers each, so they are not strong evidence for targeting decisions.

10.4 Product spending

Business Question

Which products generate the highest customer spending?

SQL Approach

Each product-spending column is summed across all customers and the products are ranked by total spending.

SQL Query

```
SQL
SELECT
    'Wines' AS product, SUM(mntwines) AS total_spending
FROM marketing
UNION ALL SELECT 'Fruits', SUM(mntfruits) FROM marketing
UNION ALL SELECT 'Meat', SUM(mntmeatproducts) FROM marketing
UNION ALL SELECT 'Fish', SUM(mntfishproducts) FROM marketing
UNION ALL SELECT 'Sweets', SUM(mntsweetproducts) FROM marketing
UNION ALL SELECT 'Gold', SUM(mntgoldprods) FROM marketing
ORDER BY total_spending DESC;
```

Result

Product	Total spending
Wines	680,816
Meat	373,968
Gold	98,609
Fish	84,057
Sweets	60,621
Fruits	58,917

Business Insight

Wines generate the highest total spending at 680,816, followed by meat at 373,968. Fruits have the lowest total at 58,917 in the supplied data.

10.5 Channel performance

Business Question

Which purchasing channels are performing best?

SQL Approach

The query sums purchases across web, catalog, and store channels and ranks the channels by total purchase count.

SQL Query

```
SQL
SELECT
    'Web' AS channel, SUM(numwebpurchases) AS total_purchases
FROM marketing
UNION ALL SELECT 'Catalog', SUM(numcatalogpurchases) FROM marketing
UNION ALL SELECT 'Store', SUM(numstorepurchases) FROM marketing
```

```
ORDER BY total_purchases DESC;
```

Result

Channel	Total purchases
Store	12,970
Web	9,150
Catalog	5,963

Business Insight

Store purchasing is the largest channel with 12,970 purchases, followed by web with 9,150 and catalog with 5,963. The result suggests that store activity is the strongest purchasing channel in this dataset.

10.6 Campaign response and purchases

Business Question

Are marketing campaigns successfully driving additional purchases?

SQL Approach

Customers are split into responders and non-responders and their average purchase and spending metrics are compared.

SQL Query

```
SQL
SELECT
    response,
    COUNT(*) AS customers,
    ROUND(AVG(total_purchase), 2) AS avg_purchases,
    ROUND(AVG(total_spend), 2) AS avg_spending,
    ROUND(AVG(numwebpurchases), 2) AS avg_web_purchases,
    ROUND(AVG(numstorepurchases), 2) AS avg_store_purchases
FROM marketing
GROUP BY response
ORDER BY response DESC;
```

Result

Response	Customers	Avg purchases	Avg spending	Avg web purchases	Avg store purchases
1	334	15.37	987.39	5.07	6.1
0	1906	12.04	538.93	3.91	5.74

Business Insight

Responders show higher average total purchases (15.37 vs 12.04) and average spending (987.39 vs 538.93). The analysis demonstrates a strong association, but the available data does not establish that the campaign itself caused the additional purchasing.

10.7 Customer value

Business Question

Which customers are most valuable based on spending and purchasing behavior?

SQL Approach

Customers are segmented using the same spending and purchase thresholds used in the SQL CASE expression: High Value, Medium Value, and Low Value.

SQL Query

```
SQL
SELECT
  CASE
    WHEN total_spend >= 2000 AND total_purchase >= 15 THEN 'High Value'
    WHEN total_spend >= 1000 AND total_purchase >= 10 THEN 'Medium Value'
    ELSE 'Low Value'
  END AS customer_segment,
  COUNT(*) AS customers,
  ROUND(AVG(income)::numeric, 2) AS avg_income,
  ROUND(AVG(total_spend), 2) AS avg_spend,
  ROUND(AVG(total_purchase), 2) AS avg_purchases
FROM marketing
GROUP BY customer_segment
ORDER BY avg_spend DESC;
```

Result

Segment	Customers	Avg income	Avg spend	Avg purchases
High Value	48	83,207.81	2,169.81	20.88
Medium Value	552	73,816.37	1,395.60	20.03
Low Value	1,640	44,068.57	294.19	9.77

Business Insight

The High Value segment contains 48 customers and has the highest average spending (2,169.81) and purchases (20.88). The Low Value segment contains 1,640 customers and averages 294.19 in spending and 9.77 purchases. These thresholds are business rules used in the project, not statistically optimized cutoffs.

10.8 Deal-purchase behavior

Business Question

Does discount-driven purchasing indicate an opportunity or dependency?

SQL Approach

The query groups customers by the number of deal purchases and compares purchase volume, spending, income, and response rate.

SQL Query

```
SQL
SELECT
  numdealspurchases,
  COUNT(*) AS customers,
  ROUND(AVG(total_purchase), 2) AS avg_total_purchases,
  ROUND(AVG(total_spend), 2) AS avg_spend,
  ROUND(AVG(income)::numeric, 2) AS avg_income,
  ROUND(AVG(response), 2) AS response_rate
FROM marketing
GROUP BY numdealspurchases
ORDER BY numdealspurchases DESC;
```

Result

Deal purchases	Customers	Avg purchases	Avg spend	Avg income	Response rate
0	46	16.8	1234.76	83531.0	34.78
1	970	12.31	720.34	56042.59	16.49
2	497	11.03	429.71	46870.29	9.66
3	297	12.26	474.36	45992.54	12.79
5	94	15.15	581.84	50270.98	17.02
10	5	18.0	795.4	57859.8	60.0
15	7	14.86	904.57	57369.0	0.0

Business Insight

Deal usage does not increase spending in a simple monotonic pattern. Customers with zero deal purchases have the highest average spend among the common/low-count groups, while customers with two deal purchases have lower average spend. The relationship is therefore mixed rather than clear evidence of discount dependency.

10.9 Country engagement

Business Question

Which countries show the strongest customer engagement?

SQL Approach

Countries are grouped and compared using customer count, responses, response rate, average spending, and average purchases.

SQL Query

```
SQL
SELECT
    country,
    COUNT(*) AS customers,
    SUM(response) AS responses,
    ROUND(100.0 * AVG(response), 2) AS response_rate,
    ROUND(AVG(total_spend), 2) AS avg_spend,
    ROUND(AVG(total_purchase), 2) AS avg_purchases
FROM marketing
GROUP BY country
ORDER BY response_rate DESC;
```

Result

Country	Customers	Responses	Response rate	Avg spend	Avg purchases
Mexico	3	2	66.67%	1,040.67	17.33
Spain	1,095	176	16.07%	604.77	12.40
Saudi Arabia	337	52	15.43%	626.32	12.77
Australia	160	23	14.37%	561.02	12.20
Canada	268	38	14.18%	628.85	12.90
Germany	120	17	14.17%	624.27	12.75
USA	109	13	11.93%	622.77	13.51
India	148	13	8.78%	537.06	11.72

Business Insight

Mexico has the highest observed response rate at 66.67%, but only three customers are present. Spain has the highest response rate among the larger customer groups at 16.07% and also contributes the most responses because it has 1,095 customers. India has the lowest response rate at 8.78% among the listed countries.

10.10 Targeting characteristics

Business Question

Which customer characteristics should be considered when targeting future campaigns?

SQL Approach

The query combines education, marital status, and age group, then compares response rate, average spending, and average purchases for each combination.

SQL Query

```
SQL
SELECT
```

```

education,
marital_status,
CASE
    WHEN age < 30 THEN 'Under 30'
    WHEN age BETWEEN 30 AND 49 THEN '30-49'
    WHEN age BETWEEN 50 AND 69 THEN '50-69'
    ELSE '70+'
END AS age_group,
COUNT(*) AS customers,
ROUND(100.0 * AVG(response), 2) AS response_rate,
ROUND(AVG(total_spend), 2) AS avg_spend,
ROUND(AVG(total_purchase), 2) AS avg_purchases
FROM marketing
GROUP BY education, marital_status, age_group
ORDER BY response_rate DESC;

```

Result

Education	Marital	Age group	Customers	Response rate	Avg spend	Avg purchases
PhD	Divorced	70+	2	100.00%	902.00	22.00
Master	Widow	70+	1	100.00%	1564.00	21.00
Graduation	Absurd	Under 30	1	100.00%	1216.00	21.00
PhD	Alone	30-49	1	100.00%	49.00	4.00
PhD	Single	Under 30	15	60.00%	670.60	11.27

Business Insight

The highest response-rate combinations in the raw ranking include several very small groups. Because some groups contain only one or two customers, the results should be used as descriptive signals rather than as standalone targeting rules. Broader segments with adequate sample sizes are more reliable for decision-making.

10.11 Customer tenure

Business Question

Does customer tenure affect campaign response?

SQL Approach

Enrollment year is used as a tenure proxy. The query compares customer count, responses, response rate, and average spending by customer year.

SQL Query

```

SQL
SELECT
    customer_year,
    COUNT(*) AS customers,
    SUM(response) AS responses,
    ROUND(100.0 * AVG(response), 2) AS response_rate,
    ROUND(AVG(total_spend), 2) AS avg_spend
FROM marketing
GROUP BY customer_year
ORDER BY customer_year;

```

Result

Customer year	Customers	Responses	Response rate	Avg spend
2012	494	132	26.72%	745.48
2013	1189	154	12.95%	601.70
2014	557	48	8.62%	490.65

Business Insight

Customers enrolled in 2012 have the highest response rate (26.72%) and average spend (745.48), while the 2014 cohort has the lowest response rate (8.62%) and average spend (490.65). This indicates a strong cohort-level association between enrollment year and engagement in the available data.

10.12 Family size

Business Question

Does family size influence customer spending?

SQL Approach

The derived total_childern field is used to group customers and compare average spending, purchasing, and response.

SQL Query

```
SQL
SELECT
    total_childern,
    COUNT(*) AS customers,
    ROUND(AVG(total_spend), 2) AS avg_spend,
    ROUND(AVG(total_purchase), 2) AS avg_purchases,
    ROUND(100.0 * AVG(response), 2) AS response_rate
FROM marketing
GROUP BY total_childern
ORDER BY total_childern;
```

Result

Children/teens	Customers	Avg spend	Avg purchases	Response rate
0	638	1106.03	16.42	26.49%
1	1128	472.73	12.01	10.28%
2	421	245.95	8.65	11.16%
3	53	274.60	7.89	3.77%

Business Insight

Customers with no children or teenagers have substantially higher average spending (1,106.03) and purchases (16.42) than customers with one or more. Their response rate is also higher at 26.49%. The pattern is descriptive and may reflect other customer characteristics not controlled for in the dataset.

10.13 Previous campaign engagement

Business Question

Does previous campaign engagement predict future campaign response?

SQL Approach

The number of previously accepted campaigns is grouped from 0 through 4 and compared with latest response rate and average spending.

SQL Query

```
SQL
SELECT
    campaign_acceptance,
    COUNT(*) AS customers,
    SUM(response) AS responses,
    ROUND(100.0 * AVG(response), 2) AS response_rate,
    ROUND(AVG(total_spend), 2) AS avg_spend
```

```
FROM marketing
GROUP BY campaign_acceptance
ORDER BY campaign_acceptance;
```

Result

Previous campaigns accepted	Customers	Responses	Response rate	Avg spend
0	1777	146	8.22%	477.81
1	325	101	31.08%	921.52
2	83	42	50.60%	1408.05
3	44	35	79.55%	1705.64
4	11	10	90.91%	1501.64

Business Insight

Response rate rises sharply with previous campaign acceptance: 8.22% for customers with zero prior acceptances versus 90.91% for the group with four. Average spending also rises from 477.81 to 1,501.64 across those endpoints. This is one of the strongest descriptive relationships in the analysis.

10.14 Web conversion by segment

Business Question

Does web conversion rate differ between customer segments?

SQL Approach

The query averages the engineered customer-level `web_conversion_rate` by response and education, while also comparing web visits and web purchases.

SQL Query

```
SQL
SELECT
    response,
    COUNT(*) AS customers,
    ROUND(AVG(web_conversion_rate):: numeric, 4) AS avg_web_conversion_rate,
    ROUND(AVG(numwebvisitsmonth), 2) AS avg_web_visits,
    ROUND(AVG(numwebpurchases), 2) AS avg_web_purchases
FROM marketing
GROUP BY response;

SELECT
    education,
    COUNT(*) AS customers,
    ROUND(AVG(web_conversion_rate):: numeric, 4) AS avg_web_conversion_rate
FROM marketing
GROUP BY education
ORDER BY avg_web_conversion_rate DESC;
```

Result

Response	Customers	Avg web conversion rate	Avg web visits	Avg web purchases
0	1906	1.0394	5.32	3.91
1	334	1.3906	5.29	5.07

Business Insight

Responders have a higher average customer-level web conversion rate (1.3906) than non-responders (1.0394). Among education groups, PhD has the highest average at 1.1646 and Basic the lowest at 0.2719. Because the source visit field is monthly while purchase counts cover a longer period, the feature is best interpreted as a project-defined ratio rather than a strict time-aligned conversion metric.

10.15 Newer-customer purchasing

Business Question

Do newer customers have different purchasing behavior?

SQL Approach

The query compares spending, total purchases, web purchases, and store purchases across the 2012, 2013, and 2014 enrollment cohorts.

SQL Query

```
SQL
SELECT
    customer_year,
    COUNT(*) AS customers,
    ROUND(AVG(total_spend), 2) AS avg_spend,
    ROUND(AVG(total_purchase), 2) AS avg_purchases,
    ROUND(AVG(numwebpurchases), 2) AS avg_web_purchases,
    ROUND(AVG(numstorepurchases), 2) AS avg_store_purchases
FROM marketing
GROUP BY customer_year
ORDER BY customer_year;
```

Result

Customer year	Customers	Avg spend	Avg purchases	Avg web purchases	Avg store purchases
2012	494	745.48	13.90	4.72	6.17
2013	1189	601.70	12.73	4.15	5.88
2014	557	490.65	10.91	3.37	5.26

Business Insight

The 2012 cohort has the highest average spend and purchase activity across the three cohorts, while 2014 is lowest on each listed measure. This supports using enrollment cohort as a useful segmentation dimension in the dashboard.

11. Power BI Dashboard

The supplied PBIX contains three report pages: Executive Overview, Customer & Campaign Analysis, and Product & Channel Analysis. The report uses KPI cards, charts, slicers, and a dark visual theme. The report definition was inspected directly from the PBIX file.

Page	Purpose / major visuals
Executive Overview	KPIs; Campaign Performance; Product Spending; Purchase Channel Performance; Customer Response by Education; Campaign Response Rate by Country; slicers for Country, Age Group, Customer Year, Education and Marital Status.
Customer & Campaign Analysis	KPIs; Campaign Response Rate by Age Group; Average Total Spending by Income Group; Average Spending by Number of Children; Previous Campaign Engagement vs Latest Response; Spending & Response by Customer Year; slicers.
Product & Channel Analysis	KPIs; Product Spending; Channel Performance-related views; Web Visits vs Average Purchases; Web Purchases by Campaign Response; Deals vs Average Total Purchases; Customer Value Segments; slicers.

Verified KPI cards

KPI	Verified dashboard field	Value from cleaned data
Total Customers	Total Customers	2,240
Total Spending	Total Spending	1,356,988
Total Purchases	Total Purchases	28,083
Campaign Response Rate	Campaign Response Rate	14.91%
Web Conversion Rate	Web Conversion Rate	76.83%

The dashboard's Web Conversion Rate KPI is a report-level measure based on aggregate web purchases relative to aggregate web visits. This differs from the customer-level `web_conversion_rate` feature created in Python..

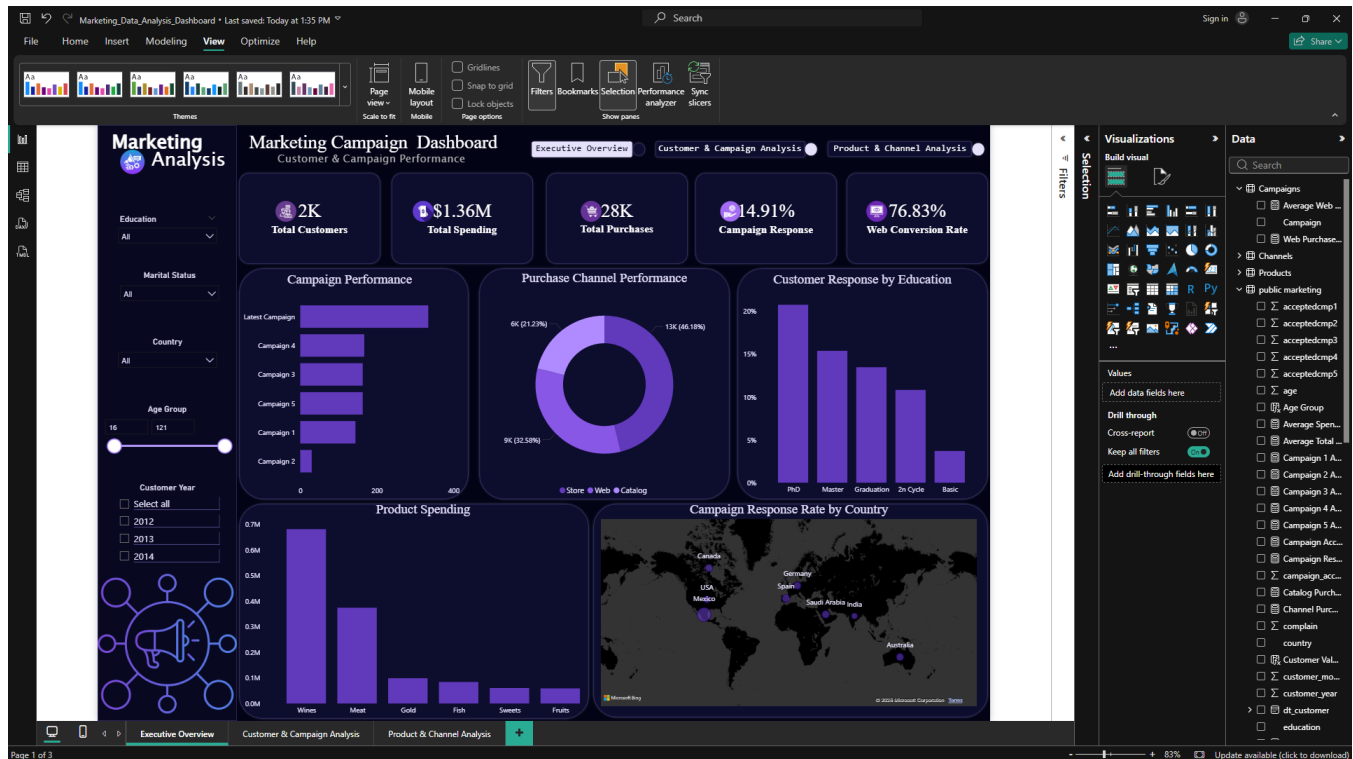
11.1. Power BI Page 1 – Executive Overview

The Executive Overview is the high-level summary page. It contains five KPI cards, five primary analytical visuals, and slicers for Country, Age Group, Customer Year, Education, and Marital Status.

Visual	Type	Main fields / measure
Campaign Performance	Clustered bar	Campaign + Campaign Accepted
Product Spending	Clustered column	Product + Product Spending
Purchase Channel Performance	Donut	Channel + Channel Purchases
Customer Response by Education	Column chart	Education + Campaign Response Rate
Campaign Response Rate by Country	Map	Country + Campaign Response Rate

Verified high-level findings from the underlying data:

- Latest Campaign has 334 accepted customers, the highest of the campaign categories in the SQL analysis.
- Wines account for the largest product spending total at 680,816.
- Store purchases are highest at 12,970, ahead of web and catalog.
- Overall campaign response rate is 14.91%.



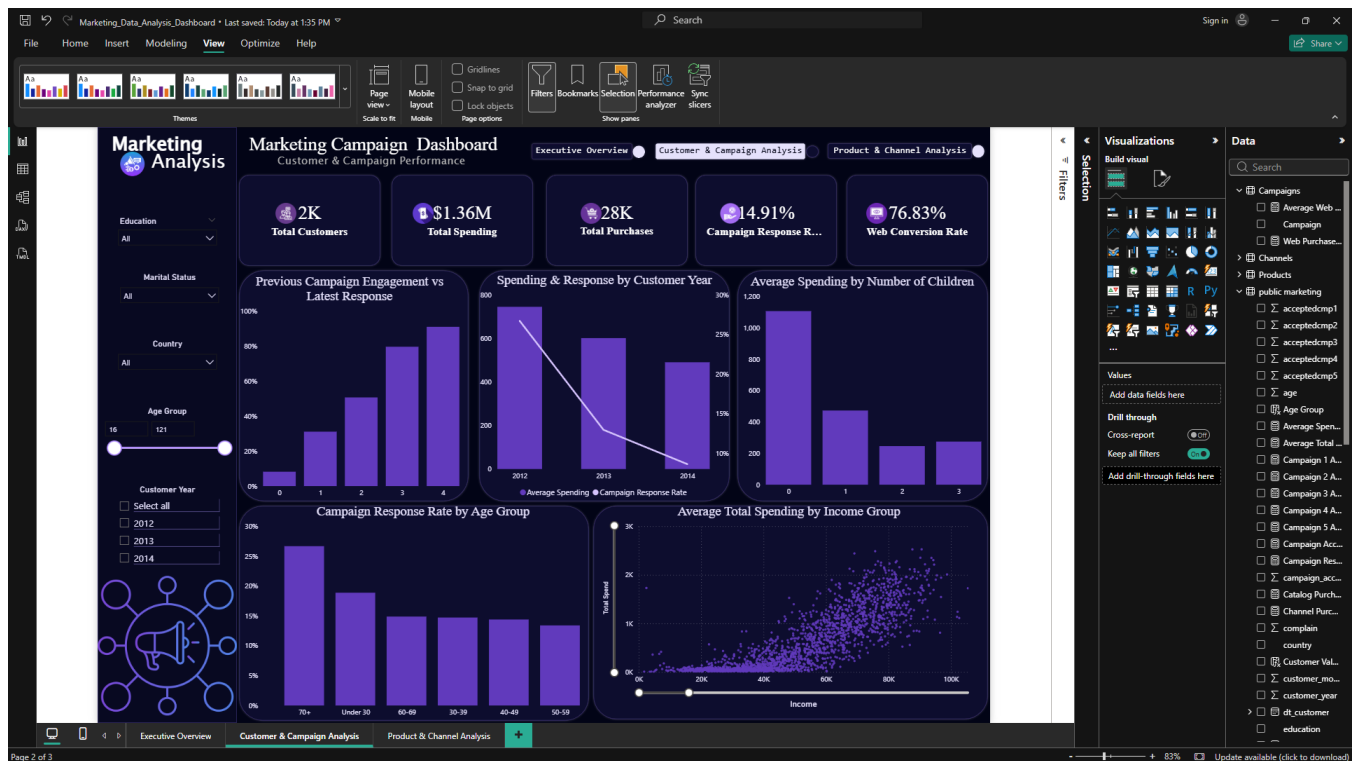
11.2. Power BI Page 2 – Customer & Campaign Analysis

This page focuses on customer characteristics and campaign behavior. The PBIX contains five KPI cards, four analytical charts, a combined spending/response chart, and slicers.

Visual	Type	Main fields / measure
Campaign Response Rate by Age Group	Clustered column	Age Group + Campaign Response Rate
Average Total Spending by Income Group	Scatter	Income + average total_spend; age and education in tooltips
Average Spending by Number of Children	Column chart	total_children + Average Spending
Previous Campaign Engagement vs Latest Response	Clustered column	campaign_acceptance + Latest Response Rate
Spending & Response by Customer Year	Line + clustered column	customer_year + Average Spending + Campaign Response Rate

The underlying SQL analysis shows:

- Responders average 987.39 in spending versus 538.93 for non-responders.
- Customers with no children/teenagers average 1,106.03 in spending, versus 472.73 for customers with one.
- Response rate rises from 8.22% for customers with no previous campaign acceptances to 90.91% for the group with four.
- The 2012 enrollment cohort has a 26.72% response rate, compared with 8.62% for 2014.



11.3. Power BI Page 3 – Product & Channel Analysis

This page focuses on purchasing behavior, online activity, deal usage, campaign response and customer value.

	Type	Main fields / measure
Product Spending	Clustered bar	Product + Product Spending
Web Visits vs Average Purchases	Column chart	numwebvisitsmonth + Average Web Purchases
Deals vs Average Total Purchases	Area chart	numdealspurchases + average total_purchase
Web Purchases by Campaign Response	Pie chart	response + sum of numwebpurchases
Customer Value Segments	Clustered bar	Customer Value Segment + Count of id

The page also includes slicers for Age Group, Country, Education, Marital Status, and Customer Year, plus the same KPI-card layer used across the report.

Customer value segmentation uses the project SQL logic:

SQL

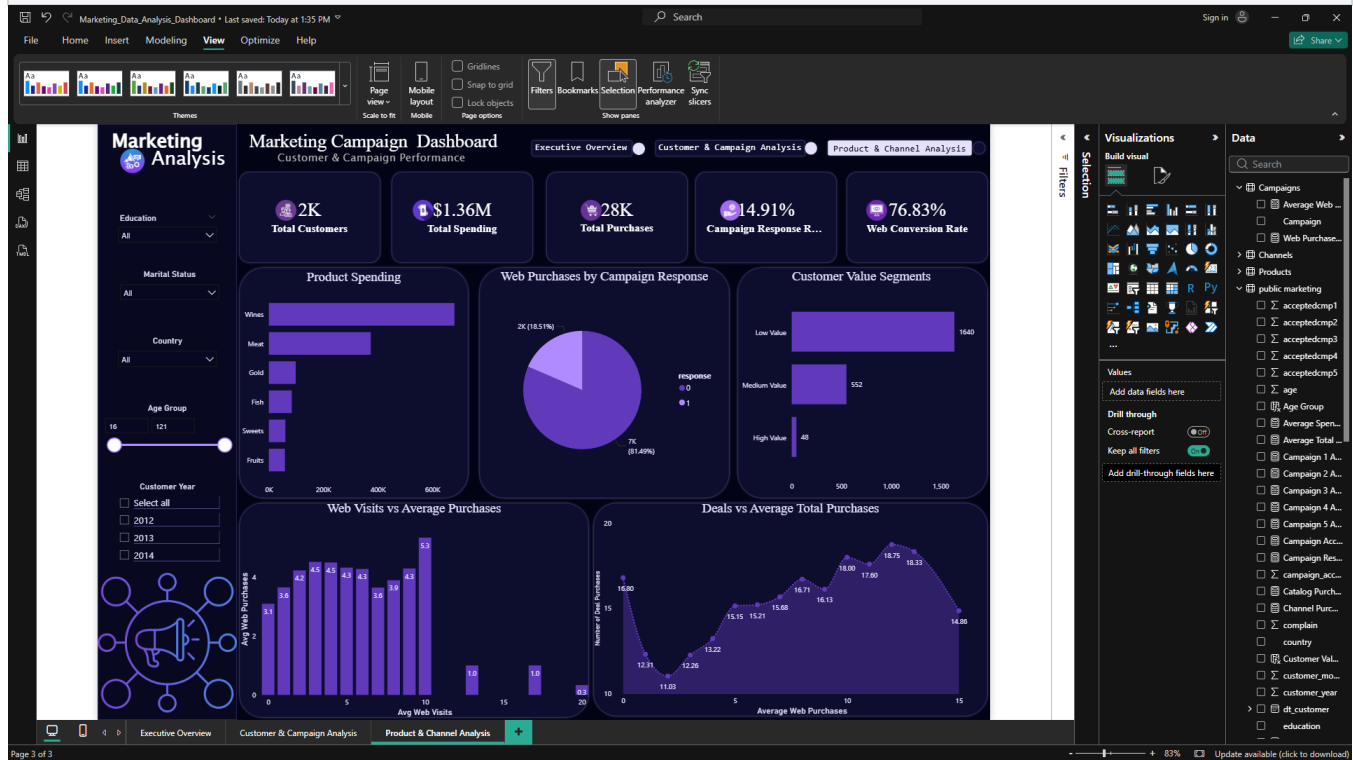
CASE

```

WHEN total_spend >= 2000 AND total_purchase >= 15
  THEN 'High Value'
WHEN total_spend >= 1000 AND total_purchase >= 10
  THEN 'Medium Value'
ELSE 'Low Value'

```

END



12. Dashboard Analysis & Insights

Observation	Meaning	Business impact
Campaign engagement	The Latest Campaign has the highest acceptance count at 334, while previous campaigns range from 30 to 167 accepted customers.	The latest campaign should be examined as a strong reference point, while the much lower Campaign 2 result warrants investigation before using it as a benchmark.
Customer response profile	Responders average 987.39 in spending and 15.37 purchases, versus 538.93 and 12.04 for non-responders.	Campaign-responsive customers are also higher-activity customers in the available data, making engagement and customer value useful dimensions for targeting.
Product mix	Wines generate 680,816 in total spending, followed by meat at 373,968.	Product-focused campaigns can prioritize the highest-spending categories while still using the dashboard to monitor smaller categories.
Channel mix	Store purchases total 12,970, web 9,150, and catalog 5,963.	Store is the strongest purchase channel by volume; web is the second-largest channel and remains important for digital engagement analysis.
Previous engagement	Latest response rate increases from 8.22% with zero previous acceptances to 90.91% with four previous acceptances.	Prior engagement is a strong descriptive signal for identifying customers who are more likely to respond again.
Customer tenure	The 2012 cohort has higher response and spending than the 2014 cohort.	Enrollment cohort can be included in campaign segmentation and retention analysis.
Family composition	Customers with no children/teenagers average 1,106.03 spending versus 472.73 for customers with one.	Household composition is associated with materially different spending patterns and can be used as a segmentation dimension.
Customer value	48 customers meet the High Value rule, 552 Medium Value, and 1,640 Low Value.	The dashboard can separate a relatively small high-value group from the broader low-value population for differentiated campaign strategies.
Geography	Spain has the largest customer base and 176 responses; Mexico has a 66.67% response rate but only three customers.	Large-country volume and response rate should be considered together; tiny groups should not drive strategic decisions alone.

13. Key Findings

Data findings

- The raw dataset contains 2,240 records and 28 columns.
- There are 24 missing income values and no duplicate rows.
- The dataset includes five education categories, eight marital-status categories and eight countries.
- The enrollment dates span 2012–2014.

Analytical findings

- Overall campaign response is 14.91%.
- The Latest Campaign has 334 accepted customers.
- Wines are the highest-spending product category at 680,816.
- Store purchases are the highest-volume channel at 12,970.
- Responders have higher average spending and purchase activity than non-responders.
- Previous campaign acceptance is strongly associated with later response in the descriptive analysis.
- The 2012 enrollment cohort has higher average spending and response than the 2014 cohort.

Business findings

- Customer value and previous engagement are useful dimensions for campaign targeting.
- Product and channel strategies should recognize the strong contribution of wines and store purchasing.
- Country-level response rates should be interpreted alongside customer counts.
- Customer segmentation should avoid over-relying on very small groups with unstable response rates.

14. Business Recommendations

Recommendation	Evidence	Action	Potential benefit
Prioritize previously engaged customers	Response rate rises from 8.22% with no previous acceptances to 90.91% with four.	Use prior campaign acceptance as a key targeting signal when designing follow-up campaigns.	Potentially improve targeting efficiency by focusing effort on customers with demonstrated engagement.
Protect and grow high-value customers	The High Value segment averages 2,169.81 spending and 20.88 purchases.	Create differentiated retention or loyalty treatment for the High Value segment while monitoring its size and behavior over time.	Supports retention of customers with the strongest observed value.
Use product and channel performance together	Wines lead spending and stores lead purchase volume.	Align product promotions with the channels customers actually use, while monitoring web behavior separately.	May improve relevance of product/channel campaigns.
Use enrollment cohort in campaign planning	2012 customers show higher response and spending than 2014 customers.	Include customer year/tenure as a segmentation variable and test whether newer customers need different onboarding or engagement strategies.	Creates a more differentiated approach to customer lifecycle management.
Treat geographic outliers cautiously	Mexico shows 66.67% response but only three customers, while Spain has 1,095 customers and 16.07% response.	Prioritize markets using both response rate and sufficient customer volume.	Reduces the risk of basing decisions on tiny samples.
Investigate discount behavior rather than assuming dependency	Deal-purchase groups show a non-monotonic relationship with spending.	Test discount campaigns against customer value and purchase outcomes rather than increasing discounts uniformly.	Helps distinguish useful promotion from unnecessary discounting.

15. Challenges & Data Limitations

- Missing income values: 24 raw income values were filled with the median (51,381.50), which is a practical preprocessing choice but can reduce natural variability.
- Web conversion metric: the customer-level web_conversion_rate uses web purchases against monthly web visits, while the purchase measure covers a longer period. It should therefore be treated as a project-defined ratio rather than a strict time-aligned conversion rate.
- Small categorical groups: some marital-status and country groups contain very few customers, producing unstable response rates.
- Customer value thresholds: the High/Medium/Low Value cutoffs are rule-based thresholds from the project SQL, not statistically optimized segmentation boundaries.
- Observational analysis: comparisons such as responders versus non-responders show associations and should not be interpreted as causal effects of campaigns.
- Excel evidence: the supplied project package does not include the Excel workbook, so detailed Excel actions could not be independently verified.
- SQL outputs: the supplied SQL file contains queries but not exported result grids. Results in this report were reproduced from the cleaned dataset using the same query logic.
- Source data scope: the dataset covers enrollment years 2012–2014 and therefore represents a historical customer period rather than current customer behavior.

16. Skills Demonstrated

Skill area	Demonstrated skills
Data exploration	Dataset inspection, descriptive statistics, categorical frequency analysis, data-quality checks
Python	Data loading, transformation, date parsing, feature engineering, validation, PostgreSQL loading
Pandas	DataFrame operations, missing-value handling, column standardization, aggregation-ready feature creation
NumPy	Numerical conditional calculation for web conversion
SQL / PostgreSQL	GROUP BY, aggregation, CASE expressions, UNION ALL, sorting, response-rate calculations, business-

	question development
Power BI	KPI cards, clustered bar/column charts, donut/pie, map, scatter, area chart, combo chart, slicers, dashboard storytelling
Analytics	Segmentation, campaign analysis, product/channel analysis, cohort analysis, customer value analysis, descriptive relationship analysis
Data storytelling	Moving from business question to cleaned data, SQL evidence, dashboard visualization and business recommendation

17. Conclusion

This project demonstrates a complete data analytics workflow from raw customer marketing data to an interactive business dashboard. The dataset was first inspected as the source for the analysis, prepared in Python using Pandas and NumPy, and validated before being loaded into PostgreSQL. SQL was then used to answer a structured set of business questions covering campaign performance, customer segments, products, channels, engagement, tenure, family composition, customer value, and online behavior.

The Power BI report turns these analytical results into three focused pages: Executive Overview, Customer & Campaign Analysis, and Product & Channel Analysis. Across the report, KPI cards and interactive slicers provide a common analytical layer while the charts allow users to compare campaign acceptance, customer response, spending, purchasing channels, engagement history and customer value.

The strongest descriptive signals include higher spending and purchasing among responders, a strong relationship between previous campaign acceptance and later response, the leading contribution of wines to total spending, the leading volume of store purchases, and meaningful differences across enrollment cohorts and household composition.

Overall, the project converts a raw customer marketing dataset into a structured analytical product that can support more informed campaign targeting, customer segmentation, product/channel planning and business discussion.

Project workflow Analytical journey: Excel → Python → Pandas/NumPy → SQL/PostgreSQL → Power BI → Business Insights → Recommendations

Appendix A – PostgreSQL Load

PYTHON

```
from sqlalchemy import create_engine

username = "postgres"
password = "<local PostgreSQL password>"
host = "localhost"
port = "5432"
database = "marketing_data"

engine = create_engine(
    f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}"
)

table_name = "marketing"

df.to_sql(
    table_name,
    engine,
    if_exists="replace",
    index=False
)
```

The supplied notebook records a successful load into table marketing. The original notebook contains a local database password; it is intentionally redacted here because credentials should not be included in portfolio documentation.

Appendix B – Screenshot Checklist

- Excel initial exploration screenshot
- Jupyter Notebook cleaning screenshot
- Jupyter Notebook analysis/validation screenshot
- SQL query + result screenshots for selected business questions
- Full-width Power BI dashboard screenshots for each report page
- Optional close-up screenshots of important Power BI visuals